Article

Aggregation Semantics for Prioritization in Cooperative Automated Decision Making

Pavel Novoa-Hernández[1,*], Mariia Godz[2] and David A. Pelta[2]

1 Dept. Computer Engineering and Systems, Universidad de La Laguna; pnovoahe@ull.edu.es
2 Dept. Computer Science and Artificial Intelligence, Universidad de Granada; {mariiagodz,dpelta}@ugr.es
* Correspondence: pnovoahe@ull.edu.es

Abstract

To be rewritten.

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1. Introduction

1.1. Research Questions and Contributions

2. Background and Related Work

2.1. Context-Aware and Cooperative Decision Support Systems

The increasing adoption of Artificial Intelligence (AI) in decision-support systems has highlighted the need to complement predictive performance with contextual, organizational, and domain-specific knowledge. In many real-world settings, decisions cannot be based solely on statistical predictions because institutional policies, expert knowledge, regulatory requirements, and operational constraints must also be considered. As a result, a growing body of research has focused on context-aware and cooperative decision-support systems capable of integrating heterogeneous sources of information into the decision-making process.

One important research direction concerns the incorporation of explicit knowledge representations alongside data-driven models. Early examples include cognitive-agent architectures that combine symbolic rules with machine-learning mechanisms to support context-aware decision making [1]. Similarly, ontology-based clinical decision-support systems have employed contextual ontologies and clinical pathways to select context-appropriate recommendations for individual cases [2]. More recent developments have extended these ideas through semantic models, knowledge graphs, and hybrid AI architectures that combine predictive models with structured domain knowledge to improve decision quality and interpretability [3,4].

A second research stream emphasizes cooperation between human and artificial intelligence. Rather than viewing decision support as a purely predictive task, these approaches consider decision making as a collaborative process in which machine-learning models provide evidence while human experts contribute contextual understanding, oversight, and governance. Cronholm and Göbel [5], for example, argue that effective decision support requires appropriate mechanisms for combining human and artificial intelligence rather than simply improving predictive accuracy. Likewise, recent surveys on human-centred AI highlight transparency, accountability, contextual awareness, and governance as fundamental requirements for trustworthy decision-support systems [6].

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Related ideas also appear in cooperative and distributed decision-making environments. Multi-agent systems, collective-intelligence frameworks, and cooperative AI architectures frequently rely on multiple decision sources that must be reconciled before a final recommendation is produced [7]. In these settings, contextual information often plays a coordinating role by constraining, guiding, or validating the outputs generated by predictive components. Similar concerns arise in governance-oriented decision-support systems, where contextual information may encode operational procedures, organizational priorities, safety requirements, or regulatory obligations [4,8].

Taken together, these studies clearly demonstrate the importance of contextual knowledge in decision-support systems. However, their primary focus typically lies in knowledge representation, ontology design, human–AI interaction, governance mechanisms, or system architecture. Once predictive evidence and contextual assessments have been generated, the numerical integration process that combines both sources of information is often treated as a secondary implementation detail. Consequently, relatively little attention has been devoted to understanding how alternative integration mechanisms affect the final recommendations produced by the system.

This limitation becomes particularly visible in architectures that explicitly separate predictive evidence from contextual assessment. Recent work on Cooperative Automated Decision-Making Systems (CADEMAS) [9] and its machine-learning instantiation CADEMAS-ML [10] exemplifies this design philosophy. CADEMAS organizes heterogeneous automated decision-making components and contextual knowledge into distinct layers, allowing predictive evidence and contextual assessments to be generated independently before being combined into a final recommendation. Such architectures make the integration mechanism itself an explicit design choice rather than an implicit implementation detail. As a result, the aggregation operator determines how predictive evidence interacts with contextual requirements, organizational priorities, or policy constraints, thereby shaping the semantics of the resulting decision process.

This observation suggests that aggregation should not be viewed merely as a numerical post-processing step. Instead, it can be interpreted as a governance mechanism that regulates the interaction between predictive and contextual information. Understanding the implications of different aggregation semantics therefore becomes essential for the design of cooperative and context-aware decision-support systems. The next subsection reviews the aggregation literature from this perspective.

2.2. Aggregation Operators and Decision Semantics

Aggregation operators constitute one of the central building blocks of multi-criteria decision making (MCDM), fuzzy decision support, and collective decision-making systems. Their purpose is to combine multiple sources of information into a single assessment while preserving, to a greater or lesser extent, the decision semantics associated with the original criteria. Consequently, aggregation has been studied extensively from both theoretical and applied perspectives.

Among the most influential aggregation frameworks are Ordered Weighted Averaging (OWA) operators [11,12], t-norm and t-conorm based operators, geometric aggregation functions, and non-additive integrals such as the Choquet and Sugeno integrals. Collectively, these approaches provide a rich mathematical vocabulary for representing different attitudes toward compensation, conjunction, disjunction, and interaction among criteria. From this perspective, the choice of aggregation operator is not merely a numerical convenience but a modeling decision that determines how favorable and unfavorable evidence may compensate for one another.

A substantial body of literature has explored these properties in traditional decision-making settings. Reviews by Mardani et al. [13] and Csiszár [14] highlight the widespread use of OWA-based aggregation and related fuzzy operators across diverse application domains. More recent studies have continued to investigate how aggregation semantics influence consensus formation, robustness, criterion interaction, and ranking behavior. Examples include consensus-oriented aggregation in group decision making [15,16], robust Choquet-based decision models [17,18], and applications employing geometric and OWA-type operators to capture alternative decision attitudes [19,20].

Despite this methodological richness, most aggregation studies are formulated within classical MCDM settings, where the aggregated inputs represent multiple criteria, attributes, preferences, or expert judgments associated with the same decision problem. The primary objective is typically to identify the most appropriate operator for combining these criteria or to develop new aggregation mechanisms with desirable mathematical properties. Comparatively less attention has been paid to architectures in which the aggregated inputs originate from fundamentally different decision layers.

This distinction is particularly important in cooperative automated decision-making systems. In architectures such as CADEMAS-ML, the aggregation process does not combine multiple criteria describing an alternative. Instead, it reconciles two conceptually distinct sources of evidence: a predictive assessment generated by machine-learning models and a contextual assessment encoding organizational priorities, regulatory requirements, expert knowledge, or policy constraints. Under this interpretation, the aggregation operator effectively determines the extent to which predictive evidence may compensate for weak contextual support, or conversely, whether contextual constraints can restrict otherwise favorable predictions. The operator therefore acts as an implicit governance mechanism rather than merely a criterion-combination tool.

For this reason, the present work focuses on three representative aggregation regimes exhibiting fundamentally different compensation semantics. The weighted arithmetic mean represents a fully compensatory integration strategy in which predictive and contextual information may offset one another. The weighted geometric mean represents a partially compensatory strategy that penalizes strong imbalances between both sources of evidence. Finally, the minimum operator represents a non-compensatory strategy in which the weakest component determines the final assessment. Together, these operators span a spectrum ranging from highly permissive to highly restrictive integration behavior while remaining interpretable and widely understood within the aggregation literature.

From the perspective of this study, the objective is therefore not to propose a new aggregation operator nor to extend aggregation theory. Instead, aggregation operators are treated as alternative decision semantics within a cooperative automated decision-making architecture. The key question is how different compensation regimes influence policy compliance, prioritization behavior, robustness to predictive perturbations, and ranking stability when predictive and contextual information must jointly determine the final recommendation.

2.3. Research Gap and Positioning

The reviewed literature reveals two largely independent research streams. On the one hand, context-aware and cooperative decision-support systems have developed increasingly sophisticated mechanisms for incorporating contextual knowledge into predictive processes through ontologies, symbolic rules, knowledge graphs, human oversight, and governance-oriented architectures [1,2,5,6]. These studies demonstrate the importance of contextual information but typically focus on knowledge representation, rule selection, system architecture, or human–AI interaction. Once predictive and contextual assessments have been generated, the mechanism responsible for their numerical integration is often treated as an implementation detail.

On the other hand, aggregation theory and fuzzy MCDM have developed a rich collection of operators with diverse compensation semantics, including OWA operators, geometric means, t-norms, Choquet integrals, and Sugeno integrals [11–14]. This literature provides extensive theoretical knowledge regarding compensation, criterion interaction, robustness, consensus formation, and preference aggregation. However, aggregation operators are generally studied as mathematical constructs or as components of multicriteria decision methods, where the aggregated inputs represent criteria, preferences, or expert judgments associated with the same decision problem.

Consequently, a gap exists between context-aware decision-support frameworks and aggregation theory. The former provides mechanisms for representing contextual knowledge but rarely investigates how different aggregation semantics affect the preservation of contextual constraints after integration. The latter offers powerful aggregation mechanisms but seldom studies their implications when heterogeneous decision signals, such as predictive machine-learning outputs and contextual assessments, must jointly determine operational recommendations.

From this perspective, the gap addressed in this work is not the absence of aggregation operators, nor the lack of mathematical knowledge about their properties. Rather, it is the lack of a systematic understanding of aggregation as a governance mechanism within cooperative automated decision-making architectures. In particular, relatively little attention has been devoted to questions such as whether contextual assessments can effectively act as veto mechanisms, how different compensation regimes influence policy compliance, how sensitive prioritization outcomes are to predictive overconfidence, or how ranking stability is affected by uncertainty in contextual information.

These observations motivate the following research questions:

- *RQ1. How do different aggregation semantics affect the interaction between predictive evidence and contextual assessments in a cooperative automated decision-making architecture?*
- *RQ2. Under what conditions can aggregation operators preserve or violate contextual constraints and veto-like policies?*
- *RQ3. How do alternative aggregation regimes influence prioritization robustness under predictive overconfidence and contextual uncertainty?*
- *RQ4. To what extent do different aggregation semantics alter the stability of the resulting rankings and intervention priorities?*

Unlike traditional fuzzy MCDM studies, the objective of the present work is not to propose a new aggregation operator, derive new algebraic results, or solve a classical multicriteria decision problem. Likewise, unlike many context-aware DSS frameworks, the focus is not on developing new mechanisms for representing contextual knowledge. Instead, aggregation operators are interpreted as alternative policy semantics within the CADEMAS architecture. By keeping both the predictive subsystem and the contextual subsystem fixed, the study isolates the aggregation mechanism as an independent design variable and examines its implications for policy compliance, robustness, and prioritization behavior. This positioning distinguishes the contribution from both aggregation theory and context-aware decision-support research.

2.4. Chronological Comparison of Related Work

Table 1 summarizes representative contributions discussed in the previous subsections and positions the present work with respect to the existing literature. The comparison focuses on four dimensions that emerge from the identified research gap: (i) the explicit integration of predictive and contextual information, (ii) the treatment of contextual con-

Table 1. Chronological comparison of representative related work. "Partial" indicates that a contribution addresses the dimension only indirectly (e.g., through rules, consensus thresholds, or architectural design) without a dedicated analysis of predictive–contextual score fusion.

Reference	Year	Main focus	Predictive + contextual integration	Policy / veto analysis	Aggregation robustness	Cooperative ADM
Teng and Tan [1]	2008	Context-aware cognitive agents	✓	Partial	No	Partial
Yao and Kumar [2]	2012	Ontology-based clinical DSS	✓	Partial	No	No
Merkert et al. [21]	2015	ML in DSS survey	Partial	No	No	No
Moral et al. [15]	2017	Consensus aggregation	No	No	Partial	No
Mardani et al. [13]	2018	Fuzzy MCDM review	No	No	No	No
Csiszár [14]	2021	OWA theory	No	Partial	Partial	No
Cronholm and Göbel [5]	2022	Human-centred AI DSS	✓	Partial	No	Partial
Yang et al. [17]	2022	Robust Choquet-based MCDM	No	Partial	✓	No
Liao et al. [18]	2023	Contextual Choquet models	Partial	No	✓	No
Trillo et al. [16]	2023	Consensus-based GDM	No	Partial	No	No
Fanfani et al. [6]	2025	Human-centred AI survey	✓	Partial	No	Partial
Present work	2026	**CADEMAS aggregation semantics**	✓	✓	✓	✓

straints or veto-like policies, (iii) the analysis of aggregation robustness, and (iv) the study of aggregation within cooperative automated decision-making architectures.

The table illustrates that prior research has typically focused either on context-aware decision-support architectures or on aggregation methods in MCDM and fuzzy decision making. Comparatively little attention has been devoted to aggregation as a governance mechanism that simultaneously integrates predictive evidence, contextual knowledge, policy constraints, and robustness considerations within a cooperative ADM framework.

3. CADEMAS and CADEMAS-ML

The Cooperative Automated Decision-Making System (CADEMAS) framework was introduced by Novoa-Hernández et al. [9] as a general architecture for context-aware decision support in which multiple heterogeneous Automated Decision-Making (ADM) units cooperate to produce a unified recommendation, score, or decision. Rather than prescribing a specific decision model, CADEMAS provides a modular structure that separates the generation of decision evidence from its contextual evaluation and final integration.

This separation is particularly relevant for the problem studied in the present work. Many decision-support systems combine predictive evidence with contextual information, but these elements are often intertwined within the same model or decision process. In contrast, CADEMAS explicitly distinguishes three conceptually different functions: (i) evidence generation, (ii) contextual assessment, and (iii) decision integration. Consequently, the aggregation mechanism can be analysed independently of both the predictive subsystem and the contextual knowledge representation.

Formally, a CADEMAS instance is represented by the tuple

$$\mathcal{S} = \langle I, U, O, K, T \rangle, \tag{1}$$

where I denotes the Input Manager, U the set of ADM units, O the Decision Integration Layer, K the Contextual Modulator, and T the universe of admissible data representations.

Given an input case, the ADM units independently generate decision evidence from the available information. In parallel, the Contextual Modulator evaluates the extent to which the same case satisfies organizational priorities, expert knowledge, regulatory requirements, or other contextual constraints. The outputs of both subsystems are then combined by the Decision Integration Layer to produce the final decision score.

An important specialization of this framework is CADEMAS-ML [10], in which the ADM units correspond to machine-learning models. In this setting, multiple predictive models cooperate to produce a consolidated predictive assessment, while contextual information is represented through a fuzzy inference system inspired by Pérez-Cañedo et al. [22] and Pérez-Cañedo et al. [23]. The contextual subsystem translates organizational priorities and institutional requirements into a normalized degree of contextual satisfaction.

To focus on the role of aggregation, the internal implementation of both subsystems is abstracted in terms of two normalized scores. For each decision case i, the predictive subsystem produces a predictive score

$$R_i \in [0, 1], \tag{2}$$

which summarizes the cooperative evidence generated by the predictive ADM units. Depending on the application domain, R_i may represent a risk estimate, a confidence score, a probability of occurrence, or any other normalized predictive assessment.

Independently, the contextual subsystem produces a context score

$$Q_i \in [0, 1], \tag{3}$$

which quantifies the degree to which the case satisfies the contextual requirements encoded by the organization, institution, or application domain.

It is worth noting that the restriction of both predictive and contextual scores to the unit interval does not limit the generality of the proposed formulation. Any continuous score defined over an interval $[a, b]$ can be transformed into the normalized domain $[0, 1]$ through an affine mapping. This transformation preserves ordering relations and therefore maintains the ranking semantics required in prioritization tasks. Consequently, R_i and Q_i should be interpreted as normalized semantic scores rather than measurements expressed in a physical unit. The use of the unit interval allows the analysis of aggregation properties independently of the original scale of the predictive and contextual information. More importantly, the lower boundary $Q_i = 0$ represents a complete *contextual violation*.

The final prioritization score is then obtained through an aggregation operator

$$P_i = A(R_i, Q_i), \tag{4}$$

where A defines the integration semantics between predictive evidence and contextual assessment. Candidate cases are subsequently ranked according to P_i, and the corresponding Top-K subset is selected for intervention or resource allocation.

It is important to emphasize that the aggregation problem considered here differs from traditional fuzzy MCDM settings. In classical MCDM, aggregation operators combine multiple criteria describing different attributes of the same alternative, typically with the objective of capturing trade-offs among decision criteria. In CADEMAS-ML, however, the operator combines heterogeneous sources of decision evidence that play fundamentally different roles: a predictive assessment generated by cooperating machine-learning models and a contextual assessment derived from organizational, regulatory, or domain-specific knowledge.

Consequently, the aggregation operator does not merely determine the relative importance of criteria. Instead, it governs how predictive evidence interacts with contextual requirements and therefore acts as an explicit policy-design component within the decision architecture. The central question is not which operator produces the most desirable aggregate score in a multicriteria sense, but rather how alternative aggregation semantics affect policy compliance, contextual veto preservation, robustness to predictive overconfidence, and ranking stability under uncertainty.

Existing CADEMAS-ML applications employ a weighted linear combination of predictive and contextual scores [10]. However, the CADEMAS framework itself does not prescribe any particular aggregation operator [9]. Consequently, different aggregation semantics can be introduced without modifying either the predictive subsystem or the contextual subsystem. This modularity provides an ideal setting for studying aggregation as a policy-design mechanism and for analysing how different compensation regimes affect prioritization, policy compliance, robustness, and ranking stability. The following section introduces the aggregation operators considered in this study and develops the theoretical properties that motivate the subsequent experimental analysis.

4. Aggregation Semantics and Theoretical Analysis

The aggregation mechanism in CADEMAS-ML determines how predictive evidence and contextual knowledge are integrated into a unified prioritization decision. Therefore, the aggregation operator should not be interpreted as a mere mathematical component, but as an explicit design choice that defines the interaction between machine-learning evidence and contextual requirements.

This section provides the theoretical foundations required to answer the research questions introduced in Section 2.3, particularly those related to the role of aggregation semantics in cooperative automated decision-making. Specifically, we analyze how different aggregation regimes modify the interaction between predictive and contextual information (RQ1), under which conditions contextual constraints and veto-like policies are preserved or violated (RQ2), how aggregation semantics influence robustness to predictive overconfidence and contextual uncertainty (RQ3), and how they affect the stability of the resulting prioritization rankings (RQ4).

The analysis does not aim to introduce a new aggregation operator or to extend classical aggregation theory. Instead, it studies existing aggregation paradigms as alternative policy semantics within the CADEMAS-ML architecture. First, we formalize the structural properties that characterize aggregation behavior, including boundedness, monotonicity, continuity, zero absorption, contextual veto preservation, and compensability. Then, we analyze three representative aggregation regimes with different decision semantics: fully compensatory linear aggregation, non-linear sub-compensatory aggregation through the weighted geometric mean, and non-compensatory minimum aggregation. Finally, we derive theoretical conditions describing contextual veto preservation, rank reversal under compensation, and sensitivity to perturbations in predictive and contextual assessments.

These results provide a formal basis for interpreting aggregation operators as policy-design components in cooperative automated decision-making systems and motivate the experimental evaluation of their practical consequences in the following sections.

4.1. Structural Properties of Aggregation Operators

Without loss of generality, all scores are assumed to be normalized within the unit interval. Any bounded continuous scoring scale $[a, b]$ can be transformed into this representation through a monotonic affine transformation, preserving the ordering relations and the aggregation semantics considered in this work.

Let

$$A : [0, 1]^2 \to [0, 1] \tag{5}$$

denote an aggregation operator that combines the predictive score R_i and the contextual score Q_i into the final prioritization score

$$P_i = A(R_i, Q_i). \tag{6}$$

Throughout this work, aggregation operators are interpreted as integration policies governing the interaction between predictive evidence and contextual knowledge within the CADEMAS-ML architecture. Rather than focusing on specific functional forms, we first introduce a set of structural properties that characterize different aggregation semantics.

Definition 1 (Boundedness)**.** *An aggregation operator is bounded if*

$$0 \leq A(r, q) \leq 1, \qquad \forall (r, q) \in [0, 1]^2. \tag{7}$$

Boundedness guarantees that the resulting prioritization score remains within the same normalized scale as the predictive and contextual assessments.

Definition 2 (Monotonicity)**.** *An aggregation operator is monotonic if*

$$r_1 \leq r_2,\ q_1 \leq q_2 \quad \Longrightarrow \quad A(r_1, q_1) \leq A(r_2, q_2). \tag{8}$$

Monotonicity ensures that improving either predictive evidence or contextual satisfaction cannot decrease the final prioritization score.

Definition 3 (Continuity)**.** *An aggregation operator is continuous if small perturbations in its inputs produce arbitrarily small changes in the aggregated score.*

Continuity is particularly relevant when predictive and contextual scores are estimated from noisy data, since it prevents abrupt changes in prioritization due to infinitesimal variations of the inputs.

Definition 4 (Zero Absorption)**.** *An aggregation operator satisfies the zero-absorption property if*

$$A(r, 0) = 0, \qquad \forall r \in [0, 1]. \tag{9}$$

Under zero absorption, a null contextual assessment completely suppresses the final prioritization score regardless of the predictive evidence available.

Definition 5 (Contextual Veto Preservation)**.** *An aggregation operator preserves contextual vetoes if*

$$Q_i = 0 \implies P_i = 0. \tag{10}$$

Within the CADEMAS-ML setting, contextual veto preservation means that contextual requirements cannot be overridden by predictive evidence. Observe that zero absorption is a sufficient condition for contextual veto preservation.

Definition 6 (Compensability)**.** *An aggregation operator is compensatory if a reduction in one input can be offset by a sufficiently large increase in the other input while maintaining or improving the aggregated score.*

Compensability determines the extent to which strong predictive evidence may compensate for weak contextual support, or vice versa. At one extreme, fully compensatory operators permit unrestricted trade-offs between prediction and context. At the opposite extreme, non-compensatory operators prevent such trade-offs and therefore behave as hard contextual constraints.

These properties provide a theoretical basis for distinguishing alternative integration semantics within CADEMAS-ML. In particular, zero absorption and compensability play a central role in determining whether contextual requirements behave as soft preferences, partial constraints, or strict vetoes. The following subsection introduces the specific aggregation operators considered in this work and analyzes how they instantiate these structural properties.

4.2. Aggregation Operators Considered

The CADEMAS framework does not prescribe a specific mechanism for combining predictive and contextual assessments [9]. Consequently, different aggregation operators can be incorporated without modifying either the predictive subsystem or the contextual subsystem.

To investigate how aggregation semantics affect prioritization behavior, policy compliance, and robustness, this work considers three representative operators that span different levels of compensability between predictive evidence and contextual requirements.

1. *Linear aggregation.* The linear operator corresponds to the integration mechanism adopted in existing CADEMAS-ML implementations [10]:

$$A_L(R_i, Q_i) = \lambda R_i + (1-\lambda) Q_i, \qquad \lambda \in [0,1]. \tag{11}$$

   The parameter λ controls the relative importance assigned to predictive and contextual information. This operator is fully compensatory because reductions in one component can always be offset by sufficient increases in the other component.
2. *Weighted geometric aggregation.* As an intermediate alternative, we consider the weighted geometric mean:

$$A_G(R_i, Q_i) = R_i^{\lambda} Q_i^{1-\lambda}, \qquad \lambda \in (0,1). \tag{12}$$

   Unlike the linear operator, the weighted geometric mean penalizes imbalanced evaluations and therefore reduces the degree of compensation between predictive and contextual information. Nevertheless, partial compensation remains possible whenever both inputs are strictly positive.
3. *Minimum aggregation.* Finally, we consider the minimum operator

$$A_M(R_i, Q_i) = \min(R_i, Q_i). \tag{13}$$

This operator represents a non-compensatory integration regime in which the final prioritization score cannot exceed the weakest component. Consequently, predictive evidence cannot compensate for insufficient contextual support.

Table 2 summarizes the structural properties of the three operators according to the definitions introduced in the previous subsection.

Table 2. Structural properties of the aggregation operators considered in this work.

Property	Linear	Geometric	Minimum
Boundedness	Yes	Yes	Yes
Monotonicity	Yes	Yes	Yes
Continuity	Yes	Yes	Yes
Zero absorption	No	Yes	Yes
Contextual veto preservation	No	Yes	Yes
Compensability	Full	Partial	No

These operators were selected because they represent three distinct integration semantics within CADEMAS-ML: unrestricted compensation, partial compensation, and strict contextual constraint. The following subsections analyze the theoretical consequences of these semantics and their implications for prioritization behavior.

4.3. Conditions for Contextual Veto Preservation

Within CADEMAS-ML, contextual information may represent different types of requirements, including organizational priorities, regulatory constraints, safety conditions, or expert-defined restrictions. In some applications, a contextual assessment should not behave as a soft preference but rather as a hard constraint. This situation occurs when a complete contextual violation must prevent an alternative from being prioritized, regardless of its predictive evidence.

This behavior can be characterized through the zero-absorption property introduced in Section 4.1. An aggregation operator provides contextual veto preservation if

$$A(r,0)=0, \qquad \forall r\in[0,1]. \tag{14}$$

Therefore, the question of whether CADEMAS-ML can enforce contextual vetoes reduces to determining whether the selected aggregation operator satisfies zero absorption.

Theorem 1 (Contextual veto preservation)**.** *Let $A:[0,1]^2\to[0,1]$ be an aggregation operator used in CADEMAS-ML. If A satisfies zero absorption, then every decision case i with a null contextual assessment,*

$$Q_i=0, \tag{15}$$

receives a null prioritization score,

$$P_i=A(R_i,Q_i)=0, \tag{16}$$

independently of its predictive score R_i.

Proof. The result follows directly from the definition of zero absorption. Since $Q_i=0$, we have $P_i=A(R_i,Q_i)=A(R_i,0)$. Because A satisfies zero absorption, $A(R_i,0)=0$, for every $R_i\in[0,1]$. Therefore, $P_i=0$. Hence, a complete contextual violation completely suppresses the final prioritization score regardless of the predictive evidence. □

Corollary 1 (Veto preservation of the considered operators)**.** *Among the aggregation operators considered in this work, the minimum and weighted geometric operators preserve contextual vetoes, whereas the linear operator does not.*

Proof. For the minimum operator, $A_M(r,0) = \min(r,0) = 0$. Similarly, for the weighted geometric operator, $A_G(r,0) = r^\lambda 0^{1-\lambda} = 0$, for $\lambda \in (0,1)$. In contrast, the linear operator yields $A_L(r,0) = \lambda r$, which is greater than zero whenever $r > 0$ and $\lambda > 0$. Therefore, predictive evidence can override a complete contextual violation under linear aggregation. □

The previous result establishes that contextual veto preservation is not a consequence of the predictive model or the contextual knowledge representation, but rather an emergent property of the integration semantics. Consequently, two CADEMAS-ML systems with identical predictive and contextual components may exhibit fundamentally different policy behavior depending only on the selected aggregation operator.

This observation motivates the subsequent analysis of rank reversal conditions and compensation thresholds, which characterize how much contextual improvement is required to offset predictive advantages under different aggregation regimes.

4.4. Conditions for Rank Reversal under Compensation

The previous section established that contextual veto preservation depends on the existence of a zero-absorption property. However, many aggregation operators used in decision-support systems follow a different philosophy: they allow partial compensation between the aggregated sources of evidence. In such cases, a sufficiently high value in one dimension may offset a lower value in another dimension.

This property is common in compensatory aggregation schemes, including weighted arithmetic means and several non-linear aggregation operators. Although compensation provides flexibility when predictive evidence and contextual information represent partially substitutable criteria, it also introduces the possibility that the final prioritization ranking differs from the ranking induced by predictive evidence alone.

To formalize this phenomenon, let $A : [0,1]^2 \to [0,1]$ be a continuous aggregation operator that is strictly increasing in both arguments. Given two alternatives i and j, assume that alternative i has stronger predictive evidence, but alternative j has better contextual alignment: $R_i > R_j$ and $Q_i < Q_j$. The final prioritization scores are then $P_i = A(R_i, Q_i)$ and $P_j = A(R_j, Q_j)$. A rank reversal occurs when $P_j > P_i$, meaning that the contextual advantage of alternative j compensates for its predictive disadvantage.

Theorem 2 (General rank reversal condition under compensation)**.** *Let $A : [0,1]^2 \to [0,1]$ be a continuous aggregation operator strictly increasing in both arguments. For two alternatives i and j satisfying predictive dominance $R_i > R_j$, assume the operator provides sufficient compensatory capacity such that there exists a contextual assessment $q_{\max} \leq 1$ satisfying $A(R_j, q_{\max}) > A(R_i, Q_i)$.*

Under these conditions, a rank reversal in favour of alternative j occurs if and only if $Q_j > Q_i + \Phi_A(R_i, R_j, Q_i)$, where Φ_A is the minimum contextual improvement required to compensate for the predictive difference, implicitly defined by $A(R_j, Q_i + \Phi_A) = A(R_i, Q_i)$.

Proof. Fix $R_i > R_j$ and Q_i. Define the difference function $g(q) = A(R_j, q) - A(R_i, Q_i)$. Because A is continuous and strictly increasing in its second argument, $g(q)$ is also continuous and strictly increasing. At $q = Q_i$, $g(Q_i) = A(R_j, Q_i) - A(R_i, Q_i) < 0$ because A is strictly increasing in its first argument and $R_j < R_i$. By hypothesis, there exists $q_{\max}$ such that $g(q_{\max}) > 0$. Therefore, by the Intermediate Value Theorem, there exists a unique contextual value $q^* \in (Q_i, q_{\max})$ such that $g(q^*) = 0$, which implies $A(R_j, q^*) = A(R_i, Q_i)$. Defining the required threshold as $\Phi_A(R_i, R_j, Q_i) = q^* - Q_i$, we obtain $P_j > P_i$ (or $g(Q_j) > 0$) if and only if $Q_j > q^*$, which is equivalent to $Q_j > Q_i + \Phi_A(R_i, R_j, Q_i)$. $\square$

The theorem shows that rank reversal is not an artefact of a particular aggregation formula, but rather a consequence of allowing compensation between predictive evidence and contextual information. The specific threshold Φ_A depends on the semantics of each aggregation operator: operators with stronger compensation require smaller contextual improvements to reverse a predictive advantage, whereas operators with reduced compensation require larger contextual differences.

This distinction is particularly relevant in CADEMAS-ML because the two aggregated components do not represent equivalent evaluation criteria. The predictive score R_i summarizes evidence generated by machine learning models, whereas the contextual score Q_i represents compliance with organizational, regulatory, or domain-specific requirements. Therefore, the choice of aggregation operator implicitly defines the degree to which contextual requirements can modify or override data-driven recommendations.

4.5. Sensitivity to Predictive and Contextual Perturbations

The previous sections analyzed the structural properties of aggregation operators in terms of compensation and contextual veto preservation. However, in practical decision-support systems, both predictive scores and contextual assessments are subject to uncertainty. Predictive models may produce unstable estimates due to limited data, model uncertainty, or distribution changes, whereas contextual assessments may vary because of incomplete knowledge, expert disagreement, or evolving organizational requirements.

Therefore, an additional relevant property of an aggregation mechanism is its sensitivity to perturbations in its input dimensions. In the context of CADEMAS-ML, this property determines how strongly changes in predictive evidence or contextual requirements affect the final prioritization score.

Let $A : [0,1]^2 \to [0,1]$ be an aggregation operator. Consider a perturbation ϵ applied to the contextual assessment while the predictive score remains fixed. The resulting variation in the prioritization score is $\Delta_Q(\epsilon) = \left|A(R, Q+\epsilon) - A(R,Q)\right|$. Similarly, a perturbation in the predictive assessment produces $\Delta_R(\epsilon) = \left|A(R+\epsilon, Q) - A(R,Q)\right|$. These quantities measure the local influence of each information source on the final decision score. Operators that strongly amplify small perturbations may lead to unstable rankings, whereas operators with bounded sensitivity provide greater robustness against uncertainty in predictive or contextual information.

Theorem 3 (Lipschitz sensitivity of aggregation operators)**.** *Let $A : [0,1]^2 \to [0,1]$ be an aggregation operator continuously differentiable in the region containing the line segment between (R,Q) and $(R+\epsilon_R, Q+\epsilon_Q)$. If there exist finite constants L_R and L_Q such that*

$$\left|\partial A/\partial R\right| \leq L_R \quad \textit{and} \qquad \left|\partial A/\partial Q\right| \leq L_Q,$$

then, for any perturbations ϵ_R and ϵ_Q,

$$\left|A(R+\epsilon_R, Q+\epsilon_Q) - A(R,Q)\right| \leq L_R|\epsilon_R| + L_Q|\epsilon_Q|.$$

Proof. By the Mean Value Theorem, there exists a point (ξ_R, ξ_Q) between (R,Q) and $(R+\epsilon_R, Q+\epsilon_Q)$ such that

$$A(R+\epsilon_R, Q+\epsilon_Q) - A(R,Q) = (\partial A/\partial R)(\xi_R, \xi_Q)\epsilon_R + (\partial A/\partial Q)(\xi_R, \xi_Q)\epsilon_Q.$$

Taking absolute values and applying the derivative bounds yields

$$\left|A(R+\epsilon_R, Q+\epsilon_Q) - A(R,Q)\right| \le L_R|\epsilon_R| + L_Q|\epsilon_Q|.$$

□

The previous theorem establishes a general robustness condition for aggregation mechanisms. The constants L_R and L_Q quantify the maximum amplification of uncertainty from each decision component. Consequently, aggregation operators with smaller sensitivity constants produce more stable prioritization scores under uncertain inputs.

For the linear aggregation operator, $A_L(R,Q) = \lambda R + (1-\lambda)Q$, the sensitivity coefficients are constant: $L_R = \lambda$ and $L_Q = 1-\lambda$. Therefore, the influence of predictive and contextual uncertainty is uniform throughout the decision space. This property explains the fully compensatory behaviour of the linear operator: a perturbation in one dimension has exactly the same marginal effect regardless of the current values of prediction and context.

For the weighted geometric operator, $A_G(R,Q) = R^{\lambda}Q^{1-\lambda}$, the local sensitivities are $\partial A_G/\partial R = \lambda R^{\lambda-1}Q^{1-\lambda}$ and $\partial A_G/\partial Q = (1-\lambda)R^{\lambda}Q^{-\lambda}$. Unlike the linear operator, sensitivity is dependent on the current state of the decision case. In particular, when Q approaches zero, contextual perturbations may have a stronger relative influence on the final score. This behaviour reflects the non-linear penalization of contextual incompatibility and provides a mathematical explanation for the stronger robustness of non-compensatory and sub-compensatory aggregation mechanisms.

For the minimum operator, $A_{\min}(R,Q) = \min(R,Q)$, the final score is completely determined by the limiting component. Consequently, the operator exhibits zero compensation: improving the non-limiting input has no effect until it becomes the active constraint. This property provides maximal preservation of contextual restrictions but may reduce responsiveness when contextual assessments are uncertain. This property can be formilzed in the following corollary:

Corollary 2 (Minimum operator stability under inactive-component perturbations)**.** *Consider a decision case evaluated under the minimum operator $A_M(R,Q) = \min(R,Q)$. If the predictive score acts as the strict limiting component ($R < Q$), the operator completely absorbs any contextual perturbation ε as long as the ordinal relationship is maintained (i.e., $R \le Q+\varepsilon$). Under these conditions, the sensitivity is zero:*

$$\left|A_M(R, Q+\varepsilon) - A_M(R,Q)\right| = |R-R| = 0.$$

Proof. Let the initial state of the decision case be defined by R and Q. By hypothesis, the predictive score is the strict limiting component, meaning $R < Q$. By the definition of the minimum operator, the initial aggregated score is:

$$A_M(R,Q) = \min(R,Q) = R.$$

Now, consider a perturbed contextual assessment $Q' = Q+\varepsilon$. We are given the condition that the perturbation ε (whether positive or negative) is bounded such that the ordinal relationship is maintained, i.e., $R \le Q+\varepsilon$. Evaluating the minimum operator under this perturbed state yields:

$$A_M(R, Q+\varepsilon) = \min(R, Q+\varepsilon) = R.$$

Calculating the absolute difference between the perturbed and initial states gives the sensitivity:

$$\left|A_M(R, Q+\varepsilon) - A_M(R, Q)\right| = |R - R| = 0.$$

Thus, the perturbation is completely absorbed, and the final prioritization score remains unaffected. $\square$

This result illustrates the asymmetric robustness of non-compensatory operators: they are insensitive to perturbations of the non-limiting component, but highly sensitive when perturbations modify the active constraint.

These analytical differences motivate the robustness experiments conducted in the following sections. Rather than evaluating aggregation operators only according to final ranking performance, we analyse how their semantic properties influence stability under perturbations of predictive evidence and contextual assessments.

5. Experimental Methodology

The theoretical analysis established that aggregation operators induce different decision semantics within CADEMAS-ML, affecting compensation, contextual constraint preservation, and sensitivity to uncertainty. However, the practical consequences of these properties emerge at the system level, where multiple decision cases compete for prioritization and intervention.

This section describes the experimental methodology designed to analyze these effects. The experiments do not aim to verify the theoretical results, but rather to investigate how different aggregation semantics translate into observable decision-making behaviors under realistic population-level scenarios. In particular, we evaluate their impact on policy compliance, robustness against predictive overconfidence, sensitivity to contextual uncertainty, and ranking stability.

The experimental design follows a Monte Carlo simulation framework in which synthetic decision populations are generated under controlled conditions. This approach allows the independent manipulation of predictive evidence, contextual assessments, uncertainty levels, and aggregation regimes while keeping the underlying decision architecture unchanged. Consequently, differences observed across scenarios can be attributed to the integration semantics rather than to variations in the predictive or contextual subsystems.

5.1. Objectives and Experimental Logic

The experimental evaluation is designed to answer how alternative aggregation semantics affect the operational behavior of CADEMAS-ML when applied to collections of competing decision cases. While the theoretical analysis characterizes individual properties of aggregation operators, the experimental analysis examines their emergent consequences when scores are used to prioritize a population of alternatives.

Specifically, the experiments focus on four aspects. First, we analyze policy compliance by measuring whether alternatives violating contextual requirements can still reach the highest-priority positions under different aggregation regimes. Second, we study the effects of predictive overconfidence, where machine-learning evidence is assumed to be systematically optimistic despite poor contextual alignment. Third, we evaluate robustness to contextual uncertainty by introducing controlled perturbations in contextual assessments. Finally, we analyze the stability of the resulting prioritization decisions through ranking and Top-K consistency measures.

For each simulated scenario, a population of decision cases is generated, where each case i is characterized by a predictive score $R_i \in [0,1]$ and a contextual score $Q_i \in [0,1]$. A given aggregation operator A produces the final prioritization score

$$P_i = A(R_i, Q_i),$$

which determines the ranking of alternatives and the selection of the Top-K candidates. The same population and perturbation conditions are evaluated under all aggregation regimes, allowing a controlled comparison of their decision semantics. The aggregation parameter λ is varied according to the specific operator definition, allowing the influence of predictive and contextual information to be systematically analyzed while preserving the normalized decision scale.

5.2. Synthetic Population Generation

The objective of the synthetic experiments is not to reproduce a particular application domain, but to create controlled decision environments in which the effects of different aggregation semantics can be isolated. Therefore, synthetic populations are generated by directly sampling the two components that define each CADEMAS-ML decision case: the predictive assessment and the contextual assessment.

Each simulated population consists of N independent decision cases. For each case i, a predictive score $R_i \in [0,1]$ and a contextual assessment $Q_i \in [0,1]$ are generated. These values represent the outputs of the predictive and contextual subsystems, respectively, while the internal mechanisms used to obtain them are abstracted away. This design allows the aggregation stage to be evaluated independently from any specific machine-learning model or knowledge representation approach.

To generate realistic score distributions, predictive and contextual values are sampled from Beta distributions:

$$R_i \sim \mathrm{Beta}(\alpha_R, \beta_R), \qquad Q_i \sim \mathrm{Beta}(\alpha_Q, \beta_Q), \tag{17}$$

which provide flexible control over concentration, dispersion, and asymmetry within the unit interval. The Beta distribution is particularly suitable for normalized confidence, risk, and satisfaction scores because it naturally represents bounded continuous assessments.

Different experimental scenarios are obtained by modifying the parameters of these distributions and by introducing controlled perturbations or dependencies between predictive and contextual assessments when required. In particular, three main scenarios are considered:

1. *Baseline decision environment.* Predictive and contextual assessments are generated independently, representing situations where machine-learning evidence and contextual knowledge provide complementary but unrelated information.
2. *Predictive overconfidence scenario.* Predictive overconfidence is modeled as a systematic upward bias in the predictive assessment. Specifically, the original predictive score is modified as

$$R_i' = \mathrm{clip}(R_i + \delta, 0, 1), \tag{18}$$

where $\delta \geq 0$ represents the magnitude of the artificial overconfidence introduced into the predictive subsystem and $\mathrm{clip}(\cdot)$ ensures that the perturbed values remain within the normalized domain. This scenario represents situations in which machine learning models assign excessively high predictive scores despite limited contextual support.

3. *Contextual uncertainty scenario.* Contextual uncertainty is simulated by introducing stochastic perturbations into the contextual assessments:

$$Q_i' = \mathrm{clip}(Q_i + \epsilon_i, 0, 1), \tag{19}$$

where

$$\epsilon_i \sim \mathcal{N}(0, \sigma_Q^2) \tag{20}$$

represents contextual uncertainty and σ_Q controls its intensity. The clipping operation guarantees that perturbed contextual assessments remain within the valid normalized range. This formulation models uncertainty arising from incomplete contextual knowledge, expert disagreement, or variations in organizational requirements.

For each scenario, identical synthetic populations are evaluated using all aggregation operators. This paired experimental design ensures that observed differences in prioritization, policy compliance, and ranking stability are attributable to the aggregation semantics rather than to variations in the generated decision cases.

All simulations are repeated using multiple random seeds to account for sampling variability. The number of generated cases, distribution parameters, perturbation levels, and random seeds are reported in the following subsection to ensure experimental reproducibility.

5.3. Monte Carlo Protocol

5.4. Evaluation Metrics

6. Simulation Results

6.1. Policy Compliance under Contextual Vetoes

6.2. Effects of Predictive Overconfidence

6.3. Robustness to Contextual Uncertainty

6.4. Sensitivity Analysis

7. Employee Attrition Case Study

7.1. Original CADEMAS-ML Application

7.2. Experimental Setup

7.3. Results and Interpretation

8. Discussion

8.1. Implications for Cooperative Automated Decision Making

8.2. Relation to Fuzzy MCDM and Aggregation Theory

8.3. Limitations

9. Conclusions and Future Work

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